



Organic solvent permeation through HKUST-1 MOF membranes: Molecular-scale insights

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ABSTRACT

HKUST-1 is a popular metal-organic framework (MOF) that is widely for organic solvent nanofiltration (OSN). This study targets a molecular-level understanding of the organic solvent permeation characteristics of HKUST-1-based OSN membranes. Non-equilibrium molecular dynamics (NEMD) simulations are conducted to investigate the permeation of two polar (namely, methanol and acetone) and five non-polar (namely, hexane, cyclohexane, toluene, xylene, and CCl₄) organic solvents through the HKUST-1 membrane. Compared to polar protic methanol and polar aprotic acetone, non-polar solvents exhibited higher solvent flux due to their weaker interactions with the HKUST-1 membrane, except for CCl₄, which has the highest viscosity and relatively larger molecular diameter. Among the non-polar solvents, the highest permeation flux exhibited by hexane ($256 \times 10^4 \pm 24 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$) is attributable to the lowest viscosity, narrow potential mean force distribution, relatively lower energy barrier, and loss of long-range interactions. Furthermore, for slow-permeating solvents, the moving wall NEMD (MW-NEMD) approach is recommended, due to its straightforward implementation and ease of use. On the other hand, for fast-permeating solvents, the boundary-driven NEMD (BD-NEMD) is more effective, as it allows for extended simulation times with reasonably sized systems, avoiding the need for large-scale setups and thereby reducing computational costs. The molecular-level insights revealed here are expected to be valuable in the advancement of MOF membranes for OSN.

1. Introduction

The interest in the organic solvent nanofiltration (OSN) process has grown significantly because of the widespread use of organic solvents, strict environmental regulations, and the increasing demand for energy efficiency. Especially, OSN emerges as a promising process for removing dissolved solutes in organic solvents with a molecular weight between 100 and 200 Da [1]. OSN has a wide range of potential applications in various industries due to its ability to selectively separate and purify organic solvents. Some of the key potential applications include the chemical and pharmaceutical industries, wastewater treatment, the food and beverage industry, fine chemicals and specialty materials, etc. [2–4]. Compared to conventional separation processes such as adsorption, evaporation, and distillation, it offers advantages such as energy

efficiency and reduced environmental impact (by enabling solvent recycling). For efficient recovery of organic solvents, OSN membranes are designed with pore sizes in the range of 5–20 Å [1].

Polymeric membranes are ubiquitous in filtration due to their availability, relatively low cost, and ease of fabrication relative to ceramic membranes. However, structural and performance challenges, especially swelling in organic solvents, limit their use in OSN. To address these issues, recent studies have focused on strategies such as chemical cross-linking [5], surface modification [6], and the incorporation of nanomaterials like graphene oxide [7], which have significantly enhanced the chemical resistance and stability of polymeric membranes. Despite these advancements, it remains difficult to synergize solvent resistance, long-term durability, and reliable separation performance in harsh conditions. Consequently, attention has turned to the

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development of novel membrane materials such as metal-organic frameworks (MOFs) [8–10], covalent organic frameworks (COFs) [11, 12], carbon nanotubes (CNTs) [13,14], graphene, graphene oxide [15–18], MoS₂ [19], and MXene [20], which offer promising characteristics for high-performance OSN applications, including tunable porosity and superior chemical stability.

Among the novel membrane materials, MOFs are distinguished by their high surface area, tunable pore size, and unique properties, as well as their ability to be synthesized under milder conditions than conventional nanomaterials (e.g., zeolite, silica, titanium dioxide, etc.) used in membrane technology. MOFs are a class of porous materials that consist of metal ions or clusters coordinated with organic ligands to form a three-dimensional, crystalline structure. The versatility of these organic ligands not only enhances their compatibility with polymers but also enables the fine-tuning of their properties. By varying metal nodes and ligands, a diverse family of MOFs can be synthesized. This means that their properties can be fine-tuned to design MOFs with tailored characteristics for specific applications.

Recently, a continuous MOF membrane has been produced as a pure polycrystalline layer on a support. These membranes are excellent candidates for solvent nanofiltration, as they have higher solvent resistance and lower swelling tendency in addition to their high separation performances. The size and nature of nanopores or nanochannels within the MOF layer are responsible for solvent permeation and solute rejection. For instance, continuous Zn-benzene-1,4-dicarboxylic acid (Zn (BDC)) polycrystalline membranes were prepared on chemically modified polyacrylonitrile (PAN) supports by Tham et al. [21]. The membranes maintained their good crystallinity even after being soaked in isopropyl alcohol (IPA) for one week, achieving 86 % rejection of Brilliant Blue R dye and an IPA permeance of approximately 2 LMH•bar⁻¹. Ma et al. [22] subsequently fabricated UiO-66-NH₂ polycrystalline membranes on polymeric supports through an in situ solvothermal synthesis approach. The UiO-66-NH₂ membrane has demonstrated strong chemical stability in a variety of organic solvents and an outstanding ability to reject Rose Bengal dye molecules (96.33 %), which were attributed to the combined effects of size exclusion and electrostatic repulsion mechanisms. These groundbreaking studies have paved the way for the development of more advanced MOF polycrystalline membranes that can exhibit superior separation performance and long-term stability for OSN applications.

Although groundbreaking studies have explored the OSN performance of MOF-based membranes, OSN remains a complex process due to the intricate interactions among solvent, solute, and membrane. The incorporation of MOFs adds further challenges, as factors such as pore size, pore distribution, metal type, and other structural characteristics significantly affect performance. A comprehensive understanding of the mechanistic aspects of OSN within the MOF structure is still needed. For this purpose, molecular simulation is an effective tool that can provide a deeper understanding of molecular transport across MOF nanopores. However, according to our knowledge, there is only one molecular-level discussion available on organic solvent permeation through MOF-based membranes. Wei et al. [23] evaluated three microporous crystalline zeolitic imidazolate frameworks (ZIF-25, ZIF-71, and ZIF-96) via molecular simulation approaches and unveiled the effects of pore chemistry in organic solvent permeation [23]. Rather than this work, solvent transport was identified in either nanochannels [24–27] or polymeric-free volumes [28–30] by molecular simulation approaches. Therefore, well-validated molecular simulation approaches can serve as powerful tools for elucidating the performance of MOF structures in OSN processes.

Non-equilibrium molecular dynamics (NEMD) simulations provide an effective way to model solvent transport under realistic filtration conditions. Among various NEMD techniques, the moveable-wall NEMD (MW-NEMD) method is one of the simplest, fastest, and most widely used approaches. As demonstrated in our previous study [25], it enables the investigation of molecular-level interactions between solutes,

solvents, and membrane materials, and provides reliable insights, particularly into solvents with low permeability. However, due to spatial limitations in MW-NEMD systems, the total simulation time is relatively short, making it challenging to obtain statistically meaningful data for fast-permeating solvents, in which transport events occur rapidly. To overcome this limitation, both MW-NEMD and boundary-driven NEMD (BD-NEMD) simulations were employed in the present study to evaluate solvent transport using various solvents with diverse properties. MW-NEMD is useful for slow-permeating solvents due to its simplicity and computational efficiency. On the other hand, BD-NEMD, which applies periodic boundary conditions to maintain a continuous flow, is beneficial for fast-permeating solvents. This method enables longer simulation durations, improves statistical reliability, and allows for a more accurate representation of flux [31], albeit at a higher computational cost. By combining both approaches, a more comprehensive understanding of solvent transport mechanisms through MOFs can be achieved. This underlies the current effort.

Recent studies by distinguished researchers have demonstrated that among MOF structures, HKUST-1 membranes, prepared on a polymer support through an electrochemically assisted interfacial growth strategy or via in situ growth within the pores of integrally skinned asymmetric polymer support, exhibit excellent dye rejection [32,33]. Additionally, the HKUST-1 membranes have been shown to have considerable permeance for methanol [9,34,35] and acetone [33,36,37] solvents. Therefore, we aimed to explore the organic solvent permeation performance of the HKUST-1 membrane, given its well-reported advantages, such as room-temperature fabrication and pore sizes within the nanofiltration range, which makes it one of the most promising MOF-based membranes. This study evaluates the permeation characteristics of organic solvents with varying polarities to identify factors governing transport behavior. Specifically, solvents with polar protic, polar aprotic, and non-polar properties were examined using two NEMD simulation approaches. The aim is to reveal the potential performance of HKUST-1 in OSN applications and to clarify how solvent properties (molecular size, polarity, viscosity), solvent–membrane interactions, and structural features of the MOF influence solvent transport.

2. Simulation methods

The HKUST-1 MOF structure (Fig. S1(a)) is adopted from the Cambridge Crystallographic Data Centre [38]. Initially, its superstructure is produced to observe a HKUST-1 membrane having a cross-section area of $51.3 \times 24.97 \text{ \AA}^2$ and a thickness in the z-direction of $24.97 \text{ \AA}$. Along the z-direction, solvent baths with lengths of 60 and 30 $\text{\AA}$ are located on the left and right sides of the membrane in order to represent the feed and permeate compartments, respectively. As shown in Fig. 1(a), the simulation cell consists of an HKUST-1 membrane immersed in pure solvent baths. Then, at the ends of the system, two rigid impermeable walls are placed to serve as pistons for the simulation of the nanofiltration process. The position of the feed and permeate pistons are adjusted based on pressures, P_{left} and P_{right} , respectively. Finally, a vacuum space with a z-directional length of 100 $\text{\AA}$ is placed on the permeate side to avoid interaction with the periodic image of the feed side.

The large-scale atomic/molecular massively parallel simulator (LAMMPS) package is used to investigate the solvent permeation [39]. Periodic boundary conditions are applied in all three orthogonal directions. The membrane atoms are constrained to their initial positions by using a spring with a constant of $50 \text{ kcal mol}^{-1} \cdot \text{\AA}^{-2}$. This setup minimizes lateral drifting of the membrane atoms due to the created pressure difference, maintains the atomic structure while allowing some degree of flexibility, and reduces computational time. Interactions between solvent molecules, pistons, and membrane atoms are modeled using van der Waals (vdW) and electrostatic interactions. The 12-6 Lennard-Jones (LJ) potential with a cutoff distance of 12 $\text{\AA}$ is used to describe the vdW interactions between two atoms. The

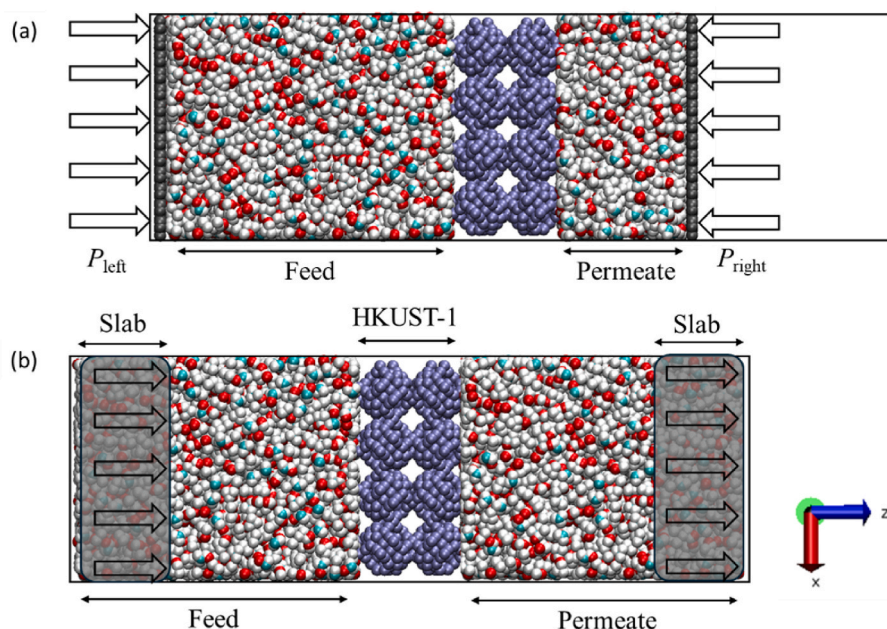


Fig. 1. (a) MW-NEMD and (b) BD-NEMD simulation systems designed for solvent nanofiltration. As an example, methanol is used as a solvent, here. The left piston is imposed with a pressure P_{left} of 400.1 MPa, while atmospheric pressure is applied on the right piston ($P_{\text{right}} = 0.1$ MPa) in MW-NEMD. Slab regions with a length of 7.5 Å from each side are defined for applying external force in BD-NEMD. Color coding of atoms: atoms of piston (grey), atoms of MOF (navy blue), oxygen (red), hydrogen (white), and carbon (cyan) of methanol. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.).

particle-particle-particle-mesh (PPPM) summation technique is employed to compute the long-range electrostatic interactions with a cutoff distance of 14 Å [40]. As proposed by Zhao et al. [41], the vdW parameters of the MOF atoms (C, O, and H) are extracted from the Dreiding force field [42] and those of Cu from the universal force field (UFF) [43]. The partial charges of the MOF atoms are adopted from an earlier study (see Fig. S1) [44]. The optimized potentials for liquid simulations-all atom (OPLS-AA) [45] force field parameters are adopted to simulate solvent molecules and paracetamol solute. Atomic charges of all organic solvents adapted to OPLS-AA force field are given in Fig. S2. The sp^2 hybridized carbon atoms of impermeable walls, which are used as pistons, are treated as neutral. The Lorentz-Berthelot mixing rules are used to estimate the atomic interactions among the different species.

The Velocity-Verlet algorithm, with a time step of 1 fs, is used to perform Molecular Dynamics (MD) simulations for the equilibration process and Non-Equilibrium MD (NEMD) simulations for the permeation calculations. Initially, each system is equilibrated for 1 ns with constant volume and temperature (NVT) ensemble using the Nosé-Hoover thermostat at 300 K, until solvent densities in both solvent baths reach the experimental values listed in Table S1. After that, P_{left} and P_{right} are adjusted to the pressures of 400.1 MPa and 0.1 MPa, respectively, to mimic the nanofiltration process using movable walls (MW)-NEMD simulations. It should be noted that the pressure gradient applied in NEMD simulations is higher than that in practical OSN systems to reduce thermal noise and improve the signal-to-noise ratio, enabling statistically reliable results within a reasonable computational time [23, 25–28]. This approach is valid for operation within the linear response regime, where flux is proportional to applied pressure. Studies have shown that this regime can persist even under perturbations comparable to intermolecular forces [46], allowing permeance values obtained under elevated pressures to be extrapolated to experimental conditions with confidence, as the underlying transport mechanism remains unchanged. All the MW-NEMD simulations are conducted for 10 ns, and data from the last 5 ns are analyzed to evaluate the permeation performance of methanol, acetone, xylene, toluene, and CCl_4 . The properties and molecular structures of all solvents used in this study are

presented in Table S1 and Fig. S1(b), respectively.

Since the length of simulation time is directly proportional to the volume of the solvent bath on the feed side, it is not easy to run a continuous MW-NEMD simulation for highly permeable solvents like hexane. If all molecules pass through the membrane, the simulation can be terminated within a short time without yielding sufficient statistical data. Therefore, a different NEMD method is followed for fast-permeating solvents, as boundary-driven (BD)-NEMD simulations [47–49], which had been applied successfully mainly for water permeation predictions. In BD-NEMD simulations, a constant external force is directly applied to fluids in the predefined region of the feed. Unlike the MW-NEMD approach, BD-NEMD allows for longer simulation times without compromising the pressure difference across the membrane, as permeated solvent molecules return to the feed bath due to periodicity. Therefore, in the system design, only impermeable walls and vacuum space are removed, and the region where the external force is applied is defined as represented in Fig. 1(b).

As for the simulation details, the simulation procedure established in our previous study [31] was utilized for the BD-NEMD simulations. Unlike the classical BD-NEMD approach, where a constant force is applied directly to individual atoms within the slab region, in this study, forces are applied to entire molecules that contain at least one atom within the slab. This modification results in more repeatable and meaningful outcomes, particularly for large solvents such as hexane. The necessity of this adjustment arises from the observation that when force is applied to atoms of large solvent molecules within the slab, atoms outside the slab exhibit different velocities, disrupting the force balance and leading to inconsistent solvent fluxes.

To ensure a more precise force distribution, the number of affected atoms at each timestep is considered. A specific $n \times f$ value is selected to achieve a pressure difference of approximately 400 MPa, similar to the MW-NEMD method, based on Equation (1):

$$n \times f = \Delta P \times A \quad (1)$$

where n represents the number of atoms in the slab region, f denotes the applied force per atom, ΔP is the pressure difference, and A is the cross-

sectional area of the MOF. This approach enables a more accurate calibration of the force-flow relationship, thereby enhancing the reliability of BD-NEMD for large solvent molecules. The other simulation parameters were adopted from our previous study [31], and all BD-NEMD simulations were conducted for 10 ns to evaluate the permeation performance of methanol, hexane, and cyclohexane. The solvent flux in both MW-NEMD and BD-NEMD simulations is defined as the number of solvent molecules transferred across the membrane from the feed to permeate reservoirs per unit area per unit time:

$$\text{Flux} = \frac{N_s M_w}{N_A A t} \quad (2)$$

where N_s , N_A , M_w , A , and t are the number of solvent molecules transferred from the feed to permeate, Avogadro constant, molecular weight of solvent, membrane cross-sectional area, and time duration, respectively.

The structures of the nanoconfined solvents within the pores of MOF membranes play a key role in determining the solvent flux. Both solvent-solvent and solvent-membrane interactions alter the packing of solvent molecules in the nanostructures, which affects the diffusion of solvent molecules through the membrane and, therefore, the solvent permeation. In this study, the solvent density distribution, radial distribution function (RDF), and potential of mean force (PMF) are investigated to understand the key influences of organic solvent nanofiltration.

The RDF describes how atom j is spatially distributed around a reference atom i , and is quantified as:

$$g_{ij}(r) = \frac{N_{ij}(r, r + \Delta r) V}{4\pi r^2 \Delta r N_i N_j} \quad (3)$$

where r is the distance between atoms i and j , $N_{ij}(r, r + \Delta r)$ is the number of j atoms located within a spherical shell of thickness Δr around atom i , V is the total system volume, and N_i and N_j are the total numbers of atoms i and j , respectively. The position and intensity of the first RDF peak provide quantitative information about interaction strength. Specifically, a peak at a lower r indicates stronger affinity between the corresponding atom types (e.g., membrane-solvent, solvent-solute, solvent-solvent).

The PMF profile represents the free energy barrier that solvent molecules must overcome to permeate through various depths of the membrane. It is calculated using the following equation [50,51]:

$$\text{PMF} = -RT \ln \left[\frac{\rho(z)}{\rho_0} \right] \quad (4)$$

where R is the gas constant, T is the temperature, $\rho(z)$ denotes the local density of solvent molecules at position z , and ρ_0 is the reference density, typically taken as the bulk solvent density.

3. Results and discussion

To ensure the reliability of the methodology of molecular simulations, it is essential to evaluate the properties of the solvent carefully by carrying out proper model parameter selection. To this end, we conducted a thorough analysis of the solvent properties and subsequently compared our findings with experimental data. For this analysis, the fluxes of three solvents, namely, methanol, acetone, and hexane, which are polar protic, polar aprotic, and non-polar, respectively, are assessed. Validations of the model are carried out in two ways. Firstly, the RDFs ($g(r)$) between O-O atom pairs of methanol and acetone molecules, as well as H-H atom pairs of hexane molecules, in the feed reservoir, are calculated, as shown in Fig. 2(a). Methanol and acetone solvents exhibit the first peaks at 2.9 and 5.5 Å, respectively, which agree with earlier studies ([52,53], respectively) and thus validate the model. Secondly, the densities of the solvents are evaluated. Under atmospheric pressure, the bulk solvent densities in the permeate reservoir are estimated, as presented in Table S2. These results agree with the densities reported at 300 K (Table S1) and thus further validate the simulation model used in this study.

The HKUST-1 membrane consists of pores with radii of 5.5 and 7 Å. Inside the HKUST-1 membrane, three different regions are marked in Fig. 2(b), namely, the apertures (I and III) and the cage (II). Fig. 2(c) presents the solvent density distribution along the z -direction. The density peaks near the membrane surfaces are due to strong membrane-solvent interactions. Because of the higher pressure applied on the feed, the feed solvent density is slightly higher than the permeate solvent density. Since the solvent density values in the membrane are directly proportional to the pore size of the membrane, the molecular size of each molecule was measured (Fig. S2), and the maximum molecular size calculated for each solvent molecule is given in Table S1.

Within the membrane, the highest density is reported at the cage (i.e., region II in Fig. 2(b)) due to the capacity for more solvent molecules to enter. On the other hand, at the apertures (i.e., regions I and III in Fig. 2(b)), lower density values are observed because of the lower free volume available for solvent molecules. Two observations are worth highlighting. Firstly, among the three solvents, the density of methanol inside the membrane region is the highest, as it has the lowest molecular size (3.83 Å) compared to acetone (5.3 Å) and hexane (9.21 Å), and consequently, more methanol molecules can enter the pores in the MOF structure (Fig. S3). Secondly, the hexane density around the cage region (II) exhibits a narrow peak, while acetone and methanol display broad and almost two peaks, respectively. This is tied to molecular interactions or the non-polarity of hexane, which affects the packing in confined spaces.

Fig. 3(a) illustrates the number of solvent molecules (N_s) transferred through the HKUST-1 membrane as a function of time simulated via MW-NEMD and BD-NEMD. Under the applied pressure gradient, a net flow of solvents from the feed reservoir to the permeate reservoir can be

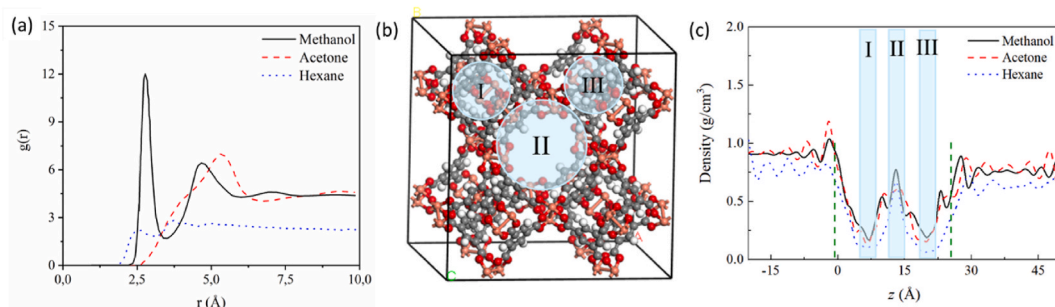


Fig. 2. (a) RDFs $g(r)$ between O-O pairs of methanol and acetone, and H-H pairs of hexane solvent molecules in the feed, (b) different regions inside the membrane, and (c) density of solvent molecules along the z -axis of the HKUST-1 membrane. The green dashed lines bound the membrane region. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

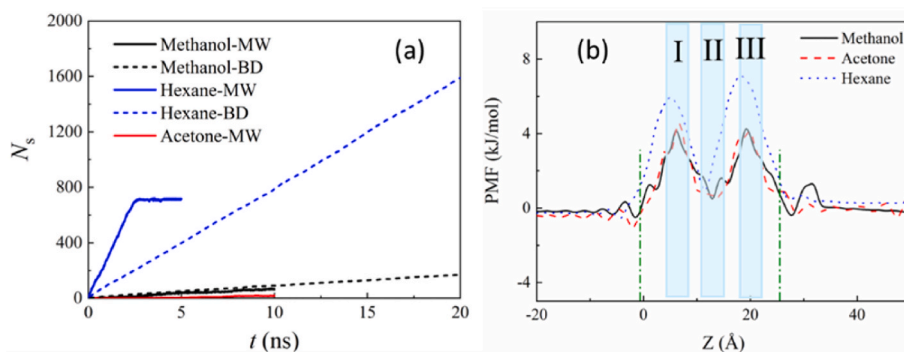


Fig. 3. Comparison of solvent polarity. (a) Number of solvent molecules permeated (N_s) as a function of simulation time simulated via MW-NEMD and BD-NEMD, and (b) PMF profiles for methanol, acetone, and hexane. The green dashed lines bound the membrane region. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

observed. Among the three solvents, the number of permeated hexane molecules obtained from MW-NEMD shows a steep linear increase with time initially and then plateaus at around 2 ns as the feed depletes. To address this, increasing the size of the feed solvent bath should be considered in MW-NEMD, which requires an expanded system size and thus increases the associated computation costs. Correspondingly, BD-NEMD, which enables the simulation of a system of the same size for an extended period, allows for the observation of the transport behavior of fast-moving solvents like hexane. Without the alteration in the density of solvent molecules defined via MW-NEMD and BD-NEMD along the z -axis of the HKUST-1 membrane (see Fig. S4), efficient solvent transport can be observed. Therefore, in the system design, only impermeable walls and vacuum space are removed, and the region where the external force is applied is as represented in Fig. 1(b). Visual representations of both system designs are available in the **video** included in **SI**, facilitating a better understanding of the simulation setup for fast-permeating hexane molecules.

Upon comparing the solvent transport calculations obtained from the MW-NEMD and BD-NEMD methods, it was observed that they both facilitated the transport of a similar number of molecules for a slow-moving solvent like methanol within the same duration of the simulation. However, it is noted that the MW-NEMD method yielded an unreasonably high number of molecules for a fast-moving solvent like hexane without the extension of the simulation time (Fig. 3(a)). It is advisable to employ the MW-NEMD method for slow-transporting solvents due to its methodological simplicity. In contrast, the use of the BD-NEMD method is recommended for fast-moving solvents, especially for conducting extended simulation times without the need to increase the system size. From the slope of the curve, the solvent flux can be calculated using Eq. (2). The solvent flux is the highest for hexane ($590 \times 10^4 \pm 164 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$ via MW-NEMD and $256 \times 10^4 \pm 24 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$ via BD-NEMD), followed by methanol ($10.9 \times 10^4 \pm 3.2 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$ via BD-NEMD and $7.2 \times 10^4 \pm 0.6 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$ via MW-NEMD) then acetone ($4.35 \times 10^4 \pm 1.5 \times 10^4 \text{ kg m}^{-2} \text{ h}^{-1}$ via MW-NEMD). This trend is not related to the size of the solvent molecules.

Fig. 3(b) displays the PMF profiles for the solvents. The PMF magnitudes are higher in the membrane region, with the peak values corresponding to the apertures (I and III in Fig. 2(b)), while the lowest PMF values in the membrane region reflect the cage (II in Fig. 2(b)). Concerning the peak magnitudes of the PMF among the solvents, the highest energy barrier is present for hexane to enter the apertures of HKUST-1. However, it displays the narrower curve in the cage (II in Fig. 2(b)), where the main transport of solvent takes place. On the contrary, acetone has the widest PMF distribution in the cage of HKUST-1, displaying the lowest flux. This suggests that the energy barrier at the entrance or in the apertures is not dominant in solvent flux for pressure-driven separation, but the width of the one in the main pore or cage is. Therefore, the interactions between the membrane and solvents

confined in the cage of membrane play a significant role in the transport of the solvent through the membrane structure. These results suggest that, for pressure-driven OSN processes, the PMF barrier at the membrane entrance does not necessarily govern the permeation rate. Instead, the solvent-membrane interactions within the main pore play a more critical role.

Significantly lower fluxes of methanol and acetone are also attributed to their polar natures, which results in a much stronger affinity to the hydrophilic HKUST-1 MOF than the non-polar hexane. These strong solvent-membrane interactions result in higher local densities within the pores but simultaneously reduce the mobility of the solvent molecules, thereby lowering their permeance. Although methanol exhibits the highest density among the studied solvents due to its small molecular size and strong affinity to the HKUST-1 framework, its permeation is hindered by this very strong interaction, which limits its transport through the membrane. When compared to acetone, which is also polar, methanol exhibits a higher permeance despite its stronger interactions, primarily because acetone has a larger molecular size that further restricts its transport through the confined pores of the membrane.

Fig. 4(a)–(c) show the RDFs between various solvent-membrane atom pairs. Clear peaks are observed for the methanol-membrane (Fig. 4(a)) and acetone-membrane (Fig. 4(b)) pairs, but not for the hexane-membrane pairs. This highlights the lack of affinity between hexane and the membrane, which underlies the high permeation flux of hexane. Therefore, the hydrophilic HKUST-1 membrane facilitates the fast permeation of non-polar solvents due to weak solvent-membrane interactions.

To further understand the permeation rate of a non-polar solvent (i. e., hexane) through the HKUST-1 membrane, a series of non-polar solvents is assessed for permeation performance. In the literature, solvent flux is often described using the combined solvent parameter $\delta_p \eta^{-1} d^{-2}$, where δ_p represents the polar term of the Hansen solubility parameter, η denotes viscosity, and d corresponds to the solvent's molecular size [23, 54,55]. However, our findings suggest that the alternative relation, $\delta^{-1} \eta^{-1} d^{-2}$, which incorporates the total Hansen solubility parameter (δ) instead of solely the polar term (δ_p), provides a stronger correlation for nonpolar solvents (hexane, cyclohexane, toluene, xylene, and carbon tetrachloride (CCl_4)), achieving an R^2 value of 0.95 (Fig. 5(a)), compared to 0.17 for the commonly used equation (Fig. S5).

Although these solvents are nonpolar, slight differences in their Hansen solubility parameters, along with significant variations in molecular size and viscosity, strongly affect their permeation behavior. Our results show that solvent flux decreases as the combined parameter increases, with viscosity identified as the dominant factor. Among the studied solvents, hexane exhibits the highest flux due to its low viscosity, whereas CCl_4 displays no permeation because of its high viscosity and relatively large molecular size. Note that fast-permeating solvents (hexane and cyclohexane) are computed via BD-NEMD, and the rest are

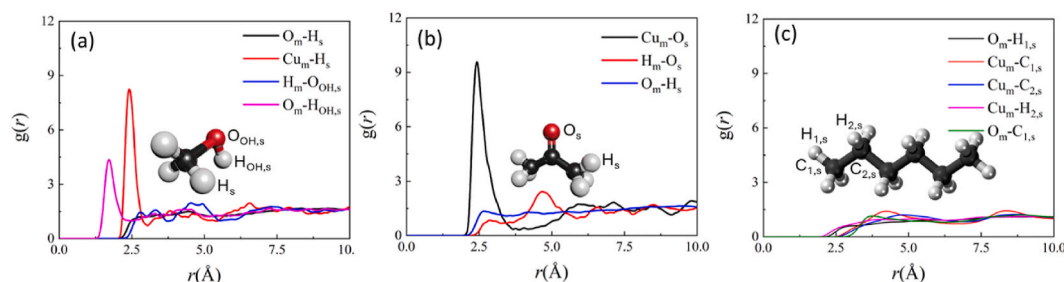


Fig. 4. RDFs ($g(r)$) for solvent-membrane atom pairs: (a) methanol, (b) acetone, and (c) hexane. The atomic MOF structure is presented in Fig. S1. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

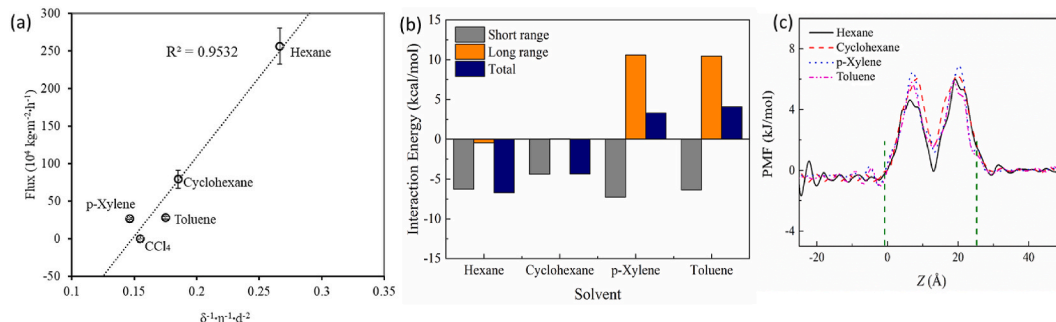


Fig. 5. Comparison of non-polar solvents. (a) solvent flux as a function of solvent combined parameter, (b) total interaction energies of solvents with membrane per solvent molecule, and (c) PMF profiles for non-polar solvents. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

examined via MW-NEMD.

Comparing the RDFs computed for all solvent molecules (see Fig. S6), toluene and xylene exhibit well-defined first coordination shells around copper atoms in HKUST-1. This observation highlights significant local atomistic interactions between these solvents and the membrane. Furthermore, the total interaction energies between the HKUST-1 membrane and the confined solvents within its pores are calculated and shown in Fig. 5(b). In the short-range interaction regime, where Coulombic and van der Waals (vdW) forces dominate, all solvents display similar interaction magnitudes. This is corroborated by similar PMF distributions for all solvents within the membrane pores, as depicted in Fig. 5(c). In contrast, long-range interactions, primarily governed by Coulombic forces, reveal a distinct trend: hexane and cyclohexane exhibit negligible long-range contributions due to their nonpolar nature, while xylene and toluene show significant contributions, likely arising from their partial polarity. These strong, long-range interactions, combined with the observed localization of p-xylene and toluene, result in low permeation through the HKUST-1 membrane. This suggests that the strong affinity of these solvents to the membrane hinders their transport. Overall, the HKUST-1 membrane is excellent for the permeation of hexane and cyclohexane because of its weak interactions with these solvents.

Finally, we sought to compare the OSN performance of HKUST-1 membrane with existing inorganic membranes in the literature to demonstrate its superior separation capabilities. In Table S3, the permeances of several MOF- and Covalent Organic Frameworks (COF)-based membranes, either free-standing or deposited on ultrafiltration membrane surface, are presented and compared with our results. COF-based membranes are included due to their structural resemblance to MOF structures and to expand the dataset for comparison. For acetone permeance, our simulation result ($13.85 \text{ LMH} \cdot \text{bar}^{-1}$) aligns well with experimental findings of HKUST-1 membrane ($16.1 \text{ LMH} \cdot \text{bar}^{-1}$ [33], $18.4 \text{ LMH} \cdot \text{bar}^{-1}$ [36], and $15.0 \text{ LMH} \cdot \text{bar}^{-1}$ [37]). Moreover, the HKUST-1 membrane outperforms other MOF/COF membranes in nonpolar solvent (especially hexane) permeation investigated either

experimentally or via simulation, likely due to its unique interaction dynamics.

To further validate the simulated methanol and hexane permeance values, a comparison involving HKUST-1 membranes in various configurations, such as functionalized forms or mixed matrix membranes, was performed, as shown in Table S3. For methanol permeance, experimental studies [9,56,57] indicate an increase in methanol permeance with MOF incorporation, reaching up to $49 \text{ LMH} \cdot \text{bar}^{-1}$ [55]. The reported values are in the same order of magnitude as the simulated performance of a free-standing MOF membrane ($34.65 \text{ LMH} \cdot \text{bar}^{-1}$). Regarding hexane permeance, HKUST-1 membranes electrochemically synthesized on PES supports using a 4V potential exhibited an extremely high permeance of $2810 \text{ LMH} \cdot \text{bar}^{-1}$, which agrees with the high hexane permeability observed in our simulations. [58]. Notably, HKUST-1 exhibits superior performance in nonpolar solvent permeation, particularly for hexane, when compared to other MOF- and COF-based membranes, studied either experimentally or computationally, likely owing to its distinct molecular interaction characteristics.

4. Conclusions

Non-equilibrium molecular dynamics (NEMD) simulations are conducted to investigate organic solvent permeation through the HKUST-1 MOF membrane. Two polar (namely, methanol and acetone) and five non-polar (namely, hexane, cyclohexane, toluene, xylene, and CCl_4) organic solvents are assessed.

The results show that the HKUST-1 membrane gives the highest solvent flux for non-polar hexane, followed by polar protic methanol and polar aprotic acetone. Despite the highest PMF values for hexane at the entrance and apertures of HKUST-1 implying the highest barriers for entering the membrane, the narrowest PMF at the cage of HKUST-1 and the weakest solvent-membrane interaction for hexane enables the fastest permeation. This indicates that the hydrophilic HKUST-1 membrane facilitates the transport of non-polar solvents due to weak solvent-membrane interactions. To better understand the permeation

characteristics of non-polar solvents, four other non-polar solvents (namely, cyclohexane, p-xylene, toluene, and CCl₄) are evaluated. Whereas hexane exhibits the highest solvent flux, no CCl₄ molecule permeates. These trends correlate well ($R^2 = 0.95$) with the total solvent-membrane interactions and the combined solvent parameter, indicating these as governing parameters for permeation.

The increasing need for OSN emphasizes the requirement for novel membrane materials that have the potential to address the challenges associated with traditional polymeric membranes. In this context, MOF-based membranes, such as HKUST-1, have emerged as promising alternatives. These findings on the permeation characteristics of various organic solvents through the HKUST-1 membrane are expected to contribute to the optimization and development of MOF membranes for OSN. This study also demonstrates the importance of implementing different molecular simulation approaches for a comprehensive understanding of solvent permeation through porous nanomaterials, suggesting the use of MW-NEMD for slow-permeating solvents and BD-NEMD for fast-permeating solvents.

CRediT authorship contribution statement

Tuğba Baysal: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Madhavi Dahanayaka:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Sadiye Velioglu:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Jia Wei Chew:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.memsci.2025.124328>.

Data availability

Data will be made available on request.

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